

Flt3L, a Cytokine and Growth Factor, Is Reduced in the Urine of Felines With Interstitial Cystitis

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Objectives: Feline idiopathic cystitis (FIC) is characterized by bladder pain with difficulties in urination and increased urinary frequency. It is one of the most common forms of Feline lower urinary tract disease (FLUTD). While the pathogenesis of FIC is not clear, inflammation plays a major role in its progression. We conducted a retrospective study to investigate whether cytokine levels change prior to the diagnosis of FIC, in order to identify a reliable non-invasive marker.

Methods: We assessed protein levels of inflammatory markers using ELISA in the blood and urine of cats clinically diagnosed with FIC ($n = 6$), FLUTD ($n = 3$), painful bladder ($n = 13$) as well as healthy controls ($n = 23$). Urine samples were obtained by cystocentesis. Mixed model/factorial analysis was performed by using JMP pro v15.0 with health conditions and the collection year of the samples as fixed effect and animal as random effect.

Results: Mean urinary Fms-like tyrosine kinase 3 ligand (Flt3L) was significantly reduced in all cats with interstitial cystitis compared

with healthy cats (39.9 vs 119.5 pg/ml respectively, $P < 0.001$). Interestingly, such lower levels of Flt3-L were observed in cats with cystitis at 4 to 5 years before the clinical diagnosis of cystitis. Conversely, Flt3L levels were higher in healthy cats over the same time period assessed in the study. Other cytokines including stem cell factor and IL12p40 were also significantly reduced in cats with cystitis compared with healthy cats. Flt3-L is a ligand for the receptor Flt3 that is expressed in hematopoietic progenitor cells, and is crucial for dendritic cell development and enhances lymphoid progenitor cells. Flt3L has been reported to have both pro-inflammatory and anti-inflammatory effects, but administration of Flt3L in human subjects increases the number of Treg cells and a subsequent reduction in inflammation. Surprisingly serum Flt3L level was not significantly different between cystitis and healthy cats although there was an increase in serum IL-18, a pro-inflammatory cytokine, when compared to controls (ns)

Conclusions: Our results suggest that Flt3L level in urine may serve as a potential diagnostic biomarker for FIC. Nutritional interventions including polyphenols and antioxidants aimed at increasing levels of Flt3L may be important in mitigating the progression of cystitis in cats.

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